

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-VII (NEW) EXAMINATION – WINTER 2024****Subject Code:3174201****Date:19-11-2024****Subject Name: Deep Learning: Principles and Practices****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Explain key features of Deep learning with the help of suitable example.	03
	(b) Discuss the working of Neural Network, in detail with help of suitable diagram.	04
	(c) Compare and contrast ANN and BNN, highlighting major advantages and limitations.	07
Q.2	(a) List key challenges in implementation of real world Deep learning applications.	03
	(b) Elaborate the role of Activation function in Neural Network, with the help of suitable example.	04
	(c) Discuss Feed forward Neural Network, with key features and applications.	07
	OR	
	(c) Explain Back propagation in Neural Network with its significance in detail.	07
Q.3	(a) List few applications of Artificial Neural Network (ANN) for ecommerce.	03
	(b) Explain the working of Neural Fuzzy Relational Systems, with the help of suitable example.	04
	(c) Illustrate Neural Network Based Fuzzy Logic Inference with the help of suitable example.	07
	OR	
Q.3	(a) List basic Fuzzy Logic Operators for realization of Neural Network.	03
	(b) Explain the working of Neuro Fuzzy Controllers with the help of suitable example.	04
	(c) Elaborate Neural Network Driven Fuzzy Reasoning, with the help of suitable example.	07
Q.4	(a) List few key features of Tensorflow with its application and advantages.	03
	(b) Discuss the concept of Transfer Learning with key advantages and disadvantages.	04
	(c) Explain working of Convolutional Neural Networks (CNN) with key advantages.	07
	OR	
Q.4	(a) List key applications of Natural Language Processing (NLP) with suitable example.	03
	(b) How process optimization can be achieved in Deep Learning, discuss with example.	04
	(c) Explain working of Recurrent Neural Network (RNN) with key advantages.	07
Q.5	(a) List few advantages of Radial Basis Function Network (RBFN).	03
	(b) How Multilayer Perceptron (MLP) system works, discuss in detail.	04
	(c) Discuss the architecture of Long Short-Term Memory (LSTM) system with suitable diagram.	07
	OR	
Q.5	(a) List key features of Self-organizing Map (SOM) in Deep learning.	03
	(b) How Deep Belief Network (DBN), system works, discuss in detail.	04
	(c) Discuss the design architecture of generative adversarial network (GAN) with suitable diagram.	07
